

Sanskar Pandey

+91 9372948245 | sanskar.p.work@gmail.com | GitHub | LinkedIn

Education

Vellore Institute of Technology, Chennai
B.Tech Computer Science (AI & ML), CGPA: 8.59/10

2023–2027

Experience

Defence Research and Development Organisation (DRDO)

AI/ML Intern

- Developed reinforcement learning agents using **Q-Learning** and **SARSA** for autonomous navigation and intelligent path planning in dynamic environments.
- Built an interactive visualization system using **Tkinter**, **Matplotlib**, and **OSMnx** to analyze agent behaviour on realistic map-based environments.
- Evaluated convergence, stability, and path efficiency across reinforcement learning algorithms to compare their navigation performance.

Publications

ST-GAT: A Leakage-Free Spatio-Temporal Graph Attention Framework for Robust Network Intrusion Detection

Submitted

- Proposed a hybrid GAT + BiLSTM architecture with leakage-free preprocessing.
- Achieved 99.14% Accuracy and 96.65% Macro-F1 on the CSE-CIC-IDS-2018 dataset.

Projects

AI Software Engineering Team

View Project

LangGraph, LangChain, Ollama, Pydantic

- Built a multi-agent AI software engineering platform that converts natural language software ideas into technical specifications and full-stack application scaffolds.
- Architected a LangGraph-based workflow of six collaborating AI agents with intelligent routing, shared state management, and structured Pydantic outputs for autonomous software planning and code generation.
- Automated the generation of Product Specifications, SRS, UI Design, Database Design, and React/Express.js project scaffolds using local LLM inference with Ollama (Qwen 2.5).

Packlite Packaging – B2B E-commerce Platform

Node.js, Express.js, PostgreSQL, Prisma, Next.js

View Project

- Led the end-to-end development of a B2B e-commerce platform designed to streamline packaging supply orders.
- Built a full-stack application using **Next.js**, **Node.js**, **Prisma**, and **PostgreSQL**, featuring product management, user authentication, shopping cart, and quotation workflow.
- Deployed the application on **Vercel** with a **Neon PostgreSQL** database, enabling automatic CI/CD from GitHub and a scalable cloud-hosted architecture.

Achievements

DATAQUEST 2.0 – 3rd Place (10,000 Prize)

Sep 2025

Krishi-Mitra (AI-Powered Crop Advisory Platform)

VIT Chennai (Hosted by L&T EduTech)

- Developed an AI-powered crop advisory platform to assist farmers with data-driven crop selection and market insights.
- Built machine learning models using **XGBoost (99% accuracy)** for crop suitability prediction and **CatBoost (91% accuracy)** for 90-day crop price forecasting.
- Developed Web and Android applications featuring multilingual support, interactive dashboards, and real-time weather alerts.

Certifications

IBM Certified – Building Agents with Agentic AI

View Certificate

- Completed a **45-hour IBM certification** covering **LangChain**, **LangGraph**, **CrewAI**, **AutoGen**, **Ollama**, **MCP**, **A2A**, **Agentic RAG**, **CRAG**, and **IBM watsonx Orchestrate**.

Delta – Full Stack Web Development (Apna College)

View Certificate

- Completed a comprehensive full-stack web development program covering **HTML**, **CSS**, **JavaScript**, **React.js**, **Node.js**, **Express.js**, **MongoDB**, **REST APIs**, **authentication**, and full-stack application development.

Technical Skills

Programming Languages: Python, Java, C, C++, JavaScript

AI/ML: LangChain, LangGraph, CrewAI, AutoGen, Ollama, FAISS, Agentic RAG, MCP, A2A, RAG,

AI Engineering: FastAPI, Vector Databases, Docker, AWS, Langfuse, LLMs, Embeddings, AI Agents, Evaluation & Observability

Web Technologies: HTML, CSS, React, Node.js, Express.js, MongoDB

Developer Tools: Git, GitHub, Visual Studio Code

Leadership

HR & Management Head, Red Ribbon Club, VIT Chennai

- Led recruitment, event planning and cross-functional coordination for student initiatives.